

** CHAPTER 2 — 

COUNTING DIGITS**

** SLIDE 1 — 

WHAT IS COUNTING DIGITS?**

🌟 MEET COUNTING DIGITS!

Sometimes, we need to find out how many digits are present in a number.

For example:

1234

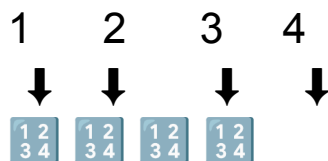
It has 4 digits.

 Simple Definition

Counting digits means finding the total number of digits present in a number.

 LOOK AT THIS!

"1234"



There are 4 digits.

So:

1234 → 4 digits

 EASY ANALOGY

Think of a number like a row of boxes 



1 2 3 4

Count the boxes:

1 2 3 4

So, there are 4 digits.

★ REMEMBER!

One box → One digit

Count the boxes → Count the digits

** SLIDE 2 —

HOW DO WE COUNT DIGITS?**

🤔 HOW CAN WE COUNT THE DIGITS?

We can keep dividing the number by 10.

Every time we divide by 10, one digit is removed.

📝 Simple Definition

Keep dividing the number by 10 until the number becomes 0.

The number of divisions tells us the number of digits.

🔍 LOOK AT THIS!

1234

$1234 \rightarrow 123 \rightarrow 12 \rightarrow 1 \rightarrow 0$

Count the divisions:

1 1234 → 123

2 123 → 12

 12 → 1

 1 → 0

So:

1234 has 4 digits.

 HOW?

We start with:

count = 0;

Then:

n = n / 10;

count++;

Repeat until n becomes 0.

 EASY ANALOGY

Think of the number as a stack of boxes 



Remove one box each time:



Keep counting how many boxes you removed.

★ REMEMBER!

/ 10 → Removes one digit 

count++ → Counts one digit 

COUNTING DIGITS USING A LOOP**

🤔 WHY DO WE USE A LOOP?

We don't know how many digits the number has.

So, we keep repeating the process until the number becomes 0.

💻 BASIC CODE

```
count = 0;
while (n != 0)
{
    n = n / 10;
    count++;
}
```

🔍 LOOK AT THIS!

For:

$n = 5678$

$5678 \rightarrow 567 \rightarrow 56 \rightarrow 5 \rightarrow 0$

And:

count = 0

count = 1

count = 2

count = 3

count = 4

So:

5678 → 4 digits

 BIG MEMORY TRICK

/ 10 → REMOVE 

count++ → COUNT 

while → REPEAT 

★ TAKEAWAY

Keep dividing by 10 and increase the count until the number becomes 0.

 QUESTIONS

1 Complete the Code

```
int n = 1234;
```

```
int count = 0;
```

```
while (n != 0)
```

```
{
```

```
    n = n _____ 10;
```

```
    count++;
```

```
}
```

 Hint: Which operator removes the last digit?

 Glossary:

/ → Division operator → divides one number by another.

2) Complete the Code

```
int n = 5678;
```

```
int count = 0;
```

```
while (n != 0)
```

```
{
```

```
    n = n / 10;
```

```
    count _____;
```

```
}
```



Hint: We need to increase the count by 1.



Glossary:

count++ → Increases the value of count by 1.

6) What happens when n becomes 0?

A) The loop continues forever

B) The loop stops

C) The number becomes 10

D) count becomes 0



Hint: Look at the condition:

```
while (n != 0)
```



Glossary:

while → Repeats a block of code while the condition is true.

4 Which statement removes one digit from a number?

A) $n = n + 10;$

B) $n = n * 10;$

C) $n = n / 10;$

D) $n = n - 10;$



Hint: Think about the operation that removes the last digit.



Glossary:

Division / → Used here to remove the last digit of an integer.

5 What is the purpose of count++?

A) Removes a digit

B) Adds 1 to the count

C) Makes the number bigger

D) Stops the loop



Hint: Think about what count means.



Glossary:

Count → The number of digits found so far.

